

Design Verification Plan and Report — VBF-T7R1NOFORCE-DVPR-01

Case t7_r1_noforce — 2026-08-25. Virtual test version. N/A items (physical nail/overcharge/crush/drop, cycle life EOL, true DFT/MD endorsement): see dvpr.md limitations list.

	Condition	Result	Verdict	Source
charge capacity	1C CC, 25 C ambient	5.0065 Ah	PASS (vs 5.0 nominal)	r5_slim_1c_dfn.json
ge temp rise	4C @45C, lumped thermal, h=100	T_max 325.54 K (+7.39 K)	PASS (< 573 K red line)	r5_slim_4c45.json
ge plating	same run, anode potential	min +0.0153 V	PASS (plated=false)	r5_slim_4c45.json
window	parameter set	2.5 – 4.2 V	informational	Chen2020
density	calc-energy caliber	491.47 Wh/kg	PASS (>= 327.18)	r5_slim_energy.json
5C 100 cyc	ec-reaction-limited aging	510.51 nm	PASS (<= 550)	r5_slim_aging45.json
W thermal runaway	t_init 325.54 K, amb 298.15 K, hA 0.531 W/K	triggered=false; T_final 317.0 K	PASS	r5_slim_nail.json
nal resistance	calc-energy	2.6415 mOhm	informational	r5_slim_energy.json